# Project MO V1
# Physical Fit-Test Plan

Eight-coupon mechanical validation program for the V1 enclosure

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| Document | Project MO V1 Physical Fit-Test Plan |
| Revision | 1.0 |
| Date | 14 September 2026 |
| Owner | Heitor de Moraes Fagundes |
| Scope | Fastener, switch, limb, USB-C, microphone, ESP32, amplifier and speaker coupons |
| Status | Prepared for supplier quotation and physical testing |

Important: Keep the editable CAD and exported 3MF files as the source package. Do not issue G-code until the selected supplier confirms the exact printer, nozzle, material profile and slicing workflow.

## Document purpose

This plan provides a repeatable method for checking whether the V1 enclosure interfaces fit their real components before the full enclosure is manufactured. It separates printing defects from CAD clearance problems and creates evidence for controlled revisions.

# 1. Test controls

## 1.1 General procedure

• Test every coupon before changing the production enclosure.

• Use the real production component and its intended orientation.

• Keep electronics disconnected and unpowered during mechanical fitting.

• Apply only gentle hand pressure. Stop if a part, tab, boss or printed wall begins to bend or crack.

• Perform three insertion/removal cycles for ordinary sliding fits. Use only one removal cycle for fragile snap tabs and self-tapping threads.

• Photograph each fitted coupon from the front and side, and record the printed coupon ID.

• Do not sand, drill, file or enlarge a fit surface before the baseline result is recorded.

## 1.2 Result classifications

| Result | Definition | Action |
|---|---|---|
| PASS | Secure fit, normal insertion and removal, correct alignment. | Accept the tested interface. |
| TIGHT | Part fits but requires excessive force or visibly stresses the print. | Review clearance; test a controlled increase. |
| LOOSE | Part fits but rattles, rotates, shifts or is not retained. | Review clearance or retaining geometry. |
| FAIL | Part cannot be inserted, retained, aligned or fastened as intended. | Stop; measure and revise CAD. |
| PRINT DEFECT | Warping, blobs, layer separation or missing features make the result unreliable. | Reprint before changing CAD. |

## 1.3 Tools and materials

• All eight PETG coupons and the matching physical components

• Digital calipers (recommended resolution: 0.01 mm)

• Manual Phillips screwdriver for M2 and M3 self-tapping screws

• Known-good USB cables for connector-access checks

• Camera or phone, permanent coupon IDs and this printed record

Use a manual screwdriver only for the coupon tests. Do not use a powered driver. If print cleanup requires a sharp tool, ask an experienced adult to handle it and record the unmodified condition first.

# 2. Pre-test print inspection

Complete this inspection before inserting any component. A failed print must not be used to justify a CAD change.

| Check | Requirement | Result |
|---|---|---|
| Identity | Label and coupon ID match the intended component. | [ ] Pass [ ] Issue |
| Material | Supplier confirms the requested PETG or reports any substitution. | [ ] Pass [ ] Issue |
| Base | Base is flat, stable and free of visible warping. | [ ] Pass [ ] Issue |
| Layers | No cracks, major separation or missing sections. | [ ] Pass [ ] Issue |
| Fit surfaces | No blobs, collapsed walls or loose strands on critical surfaces. | [ ] Pass [ ] Issue |
| Openings | Holes and slots are fully open and retain their intended shape. | [ ] Pass [ ] Issue |
| Thin features | Guides, clips, bosses and circular extensions are intact. | [ ] Pass [ ] Issue |
| Engraving | Identification is readable; cosmetic imperfections are acceptable. | [ ] Pass [ ] Issue |
| Dimensions | Critical dimensions are reasonably consistent with the reference table. | [ ] Pass [ ] Issue |

## Inspection record

| Coupon ID | Material | Layer height | Nozzle | Inspection result |
|---|---|---|---|---|
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# 3. Coupon reference matrix

Dimensions below are the coupon design references. Measurements from a printed part may differ slightly because of the printer and material process.

| Coupon | Critical geometry | Primary validation |
|---|---|---|
| Fastener | M2 boss Ø7 x 4.5; pilot Ø1.6 x 4.5; clearance Ø2.2. M3 boss Ø8 x 10; pilot Ø2.6 x 8; clearance Ø3.4. | Thread formation, boss integrity and clearance holes. |
| Switch | 45 x 45 x 3 base; centered Ø20.0 opening. | Panel fit and snap-tab engagement. |
| Limb | LP 5 x 5 x 3; LS 5.6 x 5.6 through; AP 6 x 6 x 3; AS 6.4 x 6.4 through. | Leg and arm peg/socket clearances. |
| USB-C | Opening 9.60 x 3.80, R1.80; two 3 mm walls; 4.50 edge offset; 13.8 internal guide spacing; 8 mm guide depth; 0.50 chamfer. | Dedicated USB-C interface and cable access. |
| Microphone | 3 mm face; 8 mm slot, R4 end; cradle Ø14.60 inner / Ø17.60 outer; 4 mm height; 3 x 3 tab, 1 mm height. | Microphone retention and acoustic-opening alignment. |
| ESP32 | PCB 51.0 x 27.5 x 1.6; clear 51.4 x 27.9; Ø4 x 5.5 supports; support spacing 41 x 14; 10.4 guide projection; USB opening 10 x 5, R1. | Horizontal board mount and its own USB opening. |
| Amplifier | Clear area 18.4 x 17.4; three-sided guide 3 mm thick x 3 mm high; open insertion side. | MAX98357A board fit in C-shaped guide. |
| Speaker | Ø45 outer / Ø40.6 inner; full ring 10 mm; two additional 8 mm sections; side chords 19.0 outer / 16.5 inner. | 40 mm speaker retention and cradle flexibility. |

The dedicated USB-C coupon and the ESP32 USB opening are different interfaces. Do not substitute the 10 x 5 mm ESP32 opening for the 9.60 x 3.80 mm dedicated USB-C opening.

# 4. Fastener coupon

Validate the production M2 and M3 screw interfaces before using them in the enclosure.

## Reference geometry

| Feature | Nominal value |
|---|---|
| M2 pilot boss | Ø7 x 4.5 mm; Ø1.6 x 4.5 mm blind pilot |
| M2 clearance hole | Ø2.2 mm through |
| M3 pilot boss | Ø8 x 10 mm; Ø2.6 x 8 mm blind pilot |
| M3 clearance hole | Ø3.4 mm through |

## Required parts

M2 x 8 self-tapping plastic screw; M3 x 10 self-tapping plastic screw; manual screwdriver.

## Procedure

1. Inspect both bosses and confirm the pilot holes are clear.
2. Pass each screw through its clearance hole without turning it; it should not form a thread.
3. Start each screw vertically in its labelled pilot using slow manual turns.
4. Stop if torque rises sharply, the boss whitens, bends or cracks, or the screw wanders.
5. Tighten only until firmly seated. Remove once and reinstall once to assess serviceability.

## Pass criteria

- Clearance screw passes by hand with little lateral movement.
- Pilot provides moderate, consistent resistance and the screw does not spin freely.
- Boss remains intact and the screw can be reinstalled with firm retention.

## Record

Record M2 and M3 independently: insertion behavior, approximate turns, visible damage, first tightening result and reinstall result.

| Interface | Result | Observations / measured change |
|---|---|---|
| M2 clearance | | |
| M2 pilot/boss | | |
| M3 clearance | | |
| M3 pilot/boss | | |

# 5. Switch coupon

Validate the round switch opening and snap retention in a 3 mm panel.

## Reference geometry

| Feature | Nominal value |
|---|---|
| Panel | 45 x 45 x 3 mm |
| Opening | Ø20.0 mm centered |

## Required parts

intended 20 mm round switch.

## Procedure

1. Confirm the opening is clear and inspect the switch retaining tabs.
2. Align the switch in its production orientation and press evenly using finger pressure.
3. Inspect the front flange and rear retaining tabs.
4. Gently check rotation and movement. Remove once to assess serviceability.

## Pass criteria

• No tool or excessive force is needed.

• Front flange sits flat.

• Tabs engage behind the 3 mm plate and the switch does not rattle or rotate.

• Coupon and switch tabs remain undamaged.

## Record

Record insertion force qualitatively, flange seating, tab engagement, rotation, movement and removal condition.

# 6. Limb-interface coupon

Validate both leg and arm peg/socket tolerance pairs.

## Reference geometry

| Feature | Nominal value |
|---|---|
| LP | 5.0 x 5.0 x 3.0 mm leg peg |
| LS | 5.6 x 5.6 mm through leg socket |
| AP | 6.0 x 6.0 x 3.0 mm arm peg |
| AS | 6.4 x 6.4 mm through arm socket |

## Required parts

production leg; production arm; matching peg/socket references as available.

## Procedure

1. Identify LP, LS, AP and AS and inspect their corners.
2. Insert each matching interface straight, without twisting it into place.
3. Check fully seated depth, lateral play and resistance to gentle withdrawal.
4. Perform three controlled insertion/removal cycles unless damage appears.

## Pass criteria

- Each interface reaches its intended seated position.
- No cracking, severe whitening or permanent deformation.
- Fit is secure but remains removable by hand; no obvious wobble.

## Record

Record LP, LS, AP and AS separately. Note whether the peg, socket or both would require revision.

# 7. Dedicated USB-C coupon

Validate the separate USB-C interface, perpendicular-wall relationship and cable access.

## Reference geometry

| Feature | Nominal value |
|---|---|
| Opening | 9.60 x 3.80 mm, R1.80 |
| Opening offset | Nearest edge 4.50 mm from adjacent wall |
| Walls | 3.00 mm |
| Guide | 13.8 mm internal spacing; 18.8 x 1.5 mm rails; 8 mm depth |
| Entrance | 0.50 mm equal-distance chamfer |

## Required parts

dedicated USB-C module/interface; known-good USB-C cable.

## Procedure

1. Insert the module into the U-shaped guide in its production orientation.
2. Confirm that it reaches the intended stop and aligns with the perpendicular opening.
3. With power disconnected, insert the cable through the opening into the connector.
4. Check plug seating, shell clearance and cable removal without levering the module.
5. Remove and reinsert the module three times if its connector and guide remain undamaged.

## Pass criteria

• Guide accepts and locates the module without excessive force or rattle.
• Connector is centered and accessible through the opening.
• Cable seats fully without touching or lifting the surrounding wall.
• Chamfer eases insertion and shows no damage.

## Record

Record module fit, alignment in both axes, cable seating, wall contact and any required opening or guide change.

# 8. Microphone coupon

Validate the repurposed ventilation slot as the acoustic opening and the circular microphone cradle.

## Reference geometry

| Feature | Nominal value |
|---|---|
| Face plate | 3.00 mm |
| Acoustic slot | 8.00 mm wide with R4 end; locally extended through coupon edge |
| Cradle | Ø14.60 inner; 1.50 wall; Ø17.60 outer; 4.00 mm height |
| Tab | 3 x 3 mm; 1.00 mm height |
| Alignment | Cradle center shares a slot center point |

## Required parts

production microphone module.

## Procedure

1. Inspect the slot, cradle opening and retaining tab.

2. Orient the microphone so its acoustic port faces the slot and its pins/cable have clearance.
3. Press the module into the cradle using even pressure on its rigid edges, not on sensitive components.
4. Check center alignment through the slot and gentle movement in all directions.
5. Remove and reinstall up to three times if the module and tab remain undamaged.

## Pass criteria

• Module seats without damaging the cradle or electronics.

• Acoustic port aligns with the slot and remains unobstructed.

• Cradle and tab retain the module without rattle while allowing service removal.

## Record

Record insertion direction, retention, acoustic-port alignment, pin/cable clearance and any cradle-diameter change.

# 9. ESP32 coupon

Validate the horizontal ESP32 mount beneath the display together with the ESP32-specific USB opening.

## Reference geometry

| Feature | Nominal value |
|---|---|
| PCB reference | 51.0 x 27.5 x 1.6 mm |
| Clear area | 51.40 x 27.90 mm |
| Supports | Four Ø4 x 5.5 mm; centers spaced 41.0 x 14.0 mm |
| Corner guides | L profiles; 10.40 mm projection; overall references 55 x 32 mm |
| USB opening | 10.0 x 5.0 mm, R1.0, in adjacent 3 mm wall |

## Required parts

production ESP32 board; known-good USB cable matching the ESP32 connector.

## Procedure

1. Confirm all four supports and four corner guides are intact.
2. Orient the board horizontally as installed beneath the display.
3. Insert it evenly so the PCB rests against all four support faces.
4. Check that the guides contact the board edges without bending the PCB.
5. With power disconnected, insert the USB cable through the 10 x 5 opening and confirm full seating.
6. Remove and reinstall the board three times if no damage or excessive stress appears.

## Pass criteria

- PCB seats on all four supports and remains flat.
- No guide forces the PCB to bow; no significant lateral or vertical rattle.
- USB connector aligns with the opening and accepts the cable without wall contact.
- Board remains removable for service.

## Record

Record seating at each support, PCB movement, guide stress, USB alignment, cable access and any directional clearance change.

> Do not confuse this 10 x 5 mm ESP32 USB opening with the dedicated USB-C coupon opening of 9.60 x 3.80 mm.

# 10. Amplifier coupon

Validate the MAX98357A amplifier board in the three-sided guide.

## Reference geometry

| Feature | Nominal value |
|---|---|
| Clear area | 18.40 x 17.40 mm |
| Guide | 3.00 mm thick x 3.00 mm high |
| Insertion | Left side open; top, bottom and right sides guided |

## Required parts

production MAX98357A amplifier board.

## Procedure

1. Inspect the C-shaped guide and confirm the open insertion side is clear.
2. Orient the board as installed and slide it through the open side.
3. Seat it to the right-side stop without forcing components against the guide.
4. Check movement, pin clearance and removal access.
5. Perform three sliding cycles if the PCB and guide remain undamaged.

## Pass criteria

• Board reaches its intended position without excessive force.

• Guide does not overlap or contact sensitive board components.

• Board is located without significant rattle and remains removable.

## Record

Record sliding resistance, final seating, movement, component/pin clearance and proposed directional change.

# 11. Speaker coupon

Validate the 40 mm speaker diameter, retention depth and flexible side openings.

## Reference geometry

| Feature | Nominal value |
|---|---|
| Cradle diameter | Ø45.00 outer / Ø40.60 inner; 2.20 mm radial wall |
| First section | Full circular ring, 10.00 mm |
| Extensions | Upper and lower sections, additional 8.00 mm; total 18.00 mm |
| Side openings | 19.00 mm outer chord / 16.50 mm inner chord |

## Required parts

production 40 mm speaker.

## Procedure

1. Inspect the ring, both extended sections and both side openings.
2. Hold the speaker by its rigid frame; do not press the diaphragm.
3. Align it with the cradle and insert gradually while observing cradle flex.
4. Confirm the speaker reaches its intended depth and sits evenly.

5. Check gentle movement and removal without twisting or prying against the frame.

## Pass criteria

- Speaker frame enters without excessive force or permanent cradle deformation.
- Speaker seats evenly and is retained without noticeable rattle.
- Side openings flex without cracking and return to their original position.
- Diaphragm, terminals and wires remain unobstructed.

## Record

Record insertion force, seated depth, retention, cradle recovery, terminal clearance and any diameter or chord change.

# 12. Master results record

Use one row for each printed coupon. If a coupon is reprinted, add a new row and preserve the original result.

| Coupon / ID | Print inspection | Fit result | Cycles | Evidence | CAD action |
|---|---|---|---|---|---|
| Fastener | | | | | |
| Switch | | | | | |
| Limb interface | | | | | |
| USB-C | | | | | |
| Microphone | | | | | |
| ESP32 | | | | | |
| Amplifier | | | | | |
| Speaker | | | | | |

## Detailed observation sheet

| Field | Entry |
|---|---|
| Coupon / print ID | |
| Test date and tester | |
| Printer / supplier | |
| Material / color | |
| Layer height / nozzle | |
| Measured dimensions | |
| Insertion orientation | |
| Cycle 1 | |
| Cycle 2 | |
| Cycle 3 | |
| Final classification | |
| Photo filenames | |
| Recommended action | |

# 13. CAD revision rules

Apply these rules after the physical evidence is complete.

| Observation | Check first | Controlled response |
|---|---|---|
| Tight in one direction | Blobs, seam placement, elephant foot and component orientation. | Change only the affected directional clearance; begin with a small controlled adjustment. |
| Tight everywhere | Material shrinkage, printer calibration and nominal component size. | Compare printed and CAD dimensions before increasing the full profile. |
| Loose | Correct component variant and measurement method. | Reduce only the locating clearance or revise the intended retention feature. |
| Misaligned opening | Component fully seated and correctly oriented. | Move the opening or mount reference; do not enlarge automatically. |
| Cracked thin feature | Print orientation, layer bonding and supplier settings. | Distinguish process failure from insufficient feature thickness. |
| One bad copy | Compare with another print if available. | Treat as a print defect until repeatability is established. |

## Revision workflow

1. Preserve the original result, photographs and baseline dimensions.
2. Identify whether the cause is the component, print process or CAD geometry.
3. Change one functional dimension at a time. A typical first trial is 0.10-0.20 mm, but the measured evidence takes priority.
4. Save a new Fusion version and record the exact feature and reason for the change.
5. Export a newly named 3MF and reprint only the affected coupon.
6. Repeat the same test procedure and compare results.
7. Transfer a successful coupon revision to the production model only after confirmation.

Known deferred production action: apply the verified 0.50 mm guide-entry chamfer to the editable Main Body USB-C guide after the physical USB-C fit result confirms it. Regenerate the production Main Body 3MF only after that controlled update.

# 14. Supplier print record and sign-off

The supplier should slice the 3MF models for the confirmed machine. The settings below are requirements or reporting fields, not pre-generated G-code.

| Item | Requested / confirmed value |
|---|---|
| Source files | Eight individual Project MO V1 coupon 3MF files |
| Material | PETG; supplier to state manufacturer and product |
| Layer height | Target 0.20 mm unless supplier recommends and documents another value |
| Infill | Target 20% unless supplier recommends and documents another value |
| Supports | Designed for no supports in the reviewed orientations; supplier must report any exception |
| Nozzle | Supplier to confirm |
| Printer | Supplier to confirm |
| Orientation | Supplier to preserve fit-critical surfaces and document changes |
| G-code | Generated by supplier only after machine/profile confirmation |

| Final authorization | Entry |
|---|---|
| All eight tests complete | [ ] Yes [ ] No |
| Failed interfaces re-tested | [ ] Yes [ ] No [ ] Not applicable |
| Production CAD updated | [ ] Yes [ ] No [ ] Not required |
| Production interference check | [ ] Pass [ ] Fail [ ] Pending |
| Approved by | |
| Date | |
| Notes | |

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